Curriculum Vitae: Colin Davalo

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Background

Postdoc at the Giuseppe Peano Institute, Turin 2024 - 2027
Position funded by the ERC "GENERATE", supervised by Andrea Seppi.
Year 2024-2025 spent as a guest at the Institut Fourier, Grenoble.

PhD at the University of Heidelberg 2020 - 2024
Supervised by Beatrice Pozzetti, with distinction "Summa cum Laude".

Master 2: Analysis, Arithmetic, Geometry at the University of Paris Saclay 2020
Master's thesis supervised by Olivier Guichard.

Student at ENS Paris 2017 - 2021

Preparatory classes MPSI/MP* 2015 - 2017

Scientific Production

Ideal Poisson–Voronoi tessellations for Finsler metrics on symmetric spaces 2026
(with J. Maxwell Riestenberg and Anna Wienhard) *preprint*

On Einstein Structures for $\mathbf{SO_0}(p, p+1)$-Surface Group Representations 2025
(with Parker Evans) *preprint*

G_2'-Slodowy Slices I: Geometric Structures 2025
(with Parker Evans) *preprint*

Anosov deformations of Barbot representations 2025
(with Samuel Bronstein) *preprint*

Dirichlet domains for Anosov subgroups 2026
(with Max Riestenberg) *Accepted for publication in Journal für die reine und angewandte Mathematik*

Geometric structures for maximal representations and pencils 2026
Forum of Mathematics, Sigma

Nearly geodesic immersions and domains of discontinuity 2025
Geometry & Topology 29 (2025) 2391–2461

Maximal and Borel Anosov representations in $\mathbf{Sp}(4, \mathbb{R})$ 2024
Advances in Mathematics Volume 442, April 2024, 109578

Projective structures with (Quasi-)Hitchin holonomy 2024
(with Daniele Alessandrini and Qiongling Li) *Journal of the London Mathematical Society, Volume 110, Issue 4*

Theses and Dissertations

Bachelor's Thesis "Almost Periodic Functions" 2018
with Balthazar Fléchelles, supervised by Patrick Bernard

Master's Thesis "Geometric structures for representations in the Hitchin component of $\mathbf{PSL}(n, \mathbb{K})$" 2020
supervised by Olivier Guichard

PhD Thesis "Geometric structures and representations of surface groups" 2024
supervised by Beatrice Pozzetti

Teaching

Teaching Assistant for the course "A first course in Dynamical Systems" Winter Semester 2021-2022
Master's level course
Teaching Assistant for the course "Introduction to Ergodic Theory" Summer Semester 2023
Master's level course

Other Activities

Co-organization of the "Group Meeting", Heidelberg Academic Year 2021-2022

Co-organization of the working group "Real Trees and Buildings", Heidelberg Summer Semester 2022

Co-organization of the working group "Hilbert Geometries", Grenoble Academic Year 2024-2025

Courses for the preparation of mathematical olympiads 2017-2020
Courses aimed at high school students preparing for mathematical olympiads

Participation in the problem committee of the $\mathbb{TFJM}^2$ 2016-2021
Team research competition for high school students

Long Research Stays

M1 Internship at TIFR, Mumbai (4 months) 2019
Supervised by Mahan Mj.

Research Visit at the Institut Fourier, Grenoble (4 months) 2022
Supervised by Andrea Seppi.

Research Visit at UC Riverside and the University of Michigan (3 months) 2024
Supervised by Brian Collier and Dick Canary.

Research Visit at MPI, Leipzig (3 months) 2024
Supervised by Anna Wienhard

Conference Talks

Nearly Carbon Neutral Geometric Topology Conference 2022
Online talk
Description of domains of discontinuity for Anosov representations.

Meeting of the ANR project "GAPR", Nice 2023
J.A. Dieudonné Laboratory, Nice
Nearly geodesic immersions in higher rank.

International Workshop on Higher Teichmüller Theory 2024
Chern Institute, Tianjin
Maximal representations and fibrations by pencils of quadrics.

20 years of Anosov representations 2024
MPI, Leipzig
Dirichlet domains for Anosov subgroups.

Higher rank geometric structures, Higgs bundles and physics 2025
IHP, Paris
Anosov deformations of Barbot representations.

Homogeneous Dynamics and Anosov Representations 2026
SLMath, Berkeley
Geometric structures and Anosov properties for families of representations into G_2'.

Workshop on Hitchin's Moduli Space 2026
EPFL, Lausanne
Constructing geometric structures from cyclic Higgs bundles.

New Developments of Teichmüller Theory 2026
MFO, Oberwolfach
Higher rank Teichmüller "Slices".

Seminar Talks

Seminar on Spectral Theory and Geometry 2022
Institut Fourier, Grenoble
Maximal representations and Anosov Borel in $\mathrm{Sp}(4,\mathbb{R})$.

Karlsruhe-Heidelberg-Strasbourg "Geometry Day" 2022
IRMA, Strasbourg
Maximal and Borel Anosov representations in $\mathrm{Sp}(4,\mathbb{R})$.

Seminar on Algebra, Topology and Geometry 2022
J.A. Dieudonné Laboratory, Nice
Maximal representations and Borel Anosov in $\mathrm{Sp}(4,\mathbb{R})$.

Oberseminar Differentialgeometrie 2023
Max Planck Institute, Bonn
Dirichlet domains for Anosov subgroups.

UCR Geometry Seminar 2024
UCR, Riverside
Finite-sided Dirichlet domains for Anosov representations.

RTG GeomDynTop Seminar 2024
University of Michigan, Ann Arbor
Fibrations by pencils of quadrics and maximal representations.

Geometry/Topology Seminar 2024
UIC, Chicago
Finite-sided Dirichlet domains for Anosov representations.

Geometry Seminar 2024
MPI, Leipzig
Maximal representations and fibrations by pencils of quadrics.

Geometry Seminar 2024
Universität Paderborn
Finite-sided Dirichlet domains for Anosov representations.

Seminar on Spectral Theory and Geometry 2024
Institut Fourier, Grenoble
Anosov deformations of Barbot representations.

Séminaire "Géométrie et groupes discrets" 2026
IHES, Bures sur Yvette
Slices of representations of surface groups in G_2' and associated geometric structures.

Geometry and Topology Seminar 2026
Université du Luxembourg
Slices of discrete representations constructed from Higgs bundles.

Weekly Seminar 2026
MPI MiS, Leipzig
Geometric structures from cyclic Higgs bundles.

Online French-speaking seminar "Groupes et géométrie" 2026
Online talk
Construction de représentations discrètes de groupes de surface et fibrés de Higgs cycliques.

Special Seminar of the Hitchin–Ngô Lab 2026 (upcoming)
ICMAT, Madrid
Geometric structures from cyclic Higgs bundles via parallel distributions of planes.